High level notes preproject:

Hyperspectral imaging (HI)

- More wavelengths than visual light
- Reveals more information
- Can look a specific wavelengths present in different part of image
- Applications: agriculture, pollution

HI in recycling

- Used to automate recycling sorting
- Alert problems to operator
- Specim Fx17 and Gx17 cameras

Infrared vs near infrared

- Both Vibrational spectroscopies
    - Molecules will vibrate
- Different spectrums of light used
    - NIR uses larger wavelengths than IR
- Differ in
    - Needed sample size
    - Penatration of light
    - Spectral information

Sample size

- IR vibrations are fundamental vibrations and have higher occurrence
- Overtones and combination bands happen in NIR and are occur less
    - Has to due with energy and bond length
- IR absorbed more strongly
    - Small sample sizes
    - Sample prep
- NIR absorption much weaker
    - No prep needed
    - More needed
    - Longer path lengths can be used for liquids with NIR

NIR v IR

- NIR has deeper penetration
- Ideal for studying heterogeneous samples
- IR limited to surface

Spectral information

- IR for identifying functional groups

- NIR are much broader bands
    - Difficult for identifying functional groups
    - Benefit for quantifying chemical and physical parameters
    - Prediction model required

IR for identification of substances but sample prep needed and only homogenous samples

NIR for identification and quantification of chemical and physical parameters without sample prep, homogenous and heterogenous samples

What is darkcurrent?
Dark current in hyperspectral imaging is the residual electric signal or flow of electrons generated inside an image sensor or photodetector due to thermal energy, even when no light (photons) is hitting the sensor

- Important to have for calibration
    - Tells you what the sensor measures without light given the environmental conditions
    - Subtract this out of measurements

Project Idea

- Train a model to identify hazardous E Waste in hyperspectral images

Project design choices

Dataset: “Near-infrared reflection spectrum of plastics”, Youngjun; Jaemin; Sangwoo (2022), doi: 10.17632/jg2z4g6pdh.1

- *Sensor type:* SPECIM FX17e
- *Bit depth:* 12
- *Spectral bands:* 224 (total number of bands looked at from 932nm to 1720nm)
- *Spatial samples:* 640
- *Frame per second:* 200

Provides

- Wavelengths from channels 1-224 (from 932 to 1720)
- Dark current for each channel
    - Does this need to be filtered out?
- Training set with 5 class IDs
    - 1 for PE, 2 for PP, 3 for PET, 4 for PS and 5 for PVC (5 most common polymers)
        - PE = Polyethylene (plastic bottles, containers, films, bags - repeating CH2 groups gives strong C-H absorption features)
            - Widely recycled

        - PP = polypropylene (food containers, bottle caps, automotive components - presence of methyl CH3 distinguishes from PE
            - Recyclable but sorting availability varies
        - PET = Polyethylene terephthalate
            - Used for beverage bottles, polyester fibres and food packaging
                - Aromatic rings and ester groups distinguish its spectral sig
        - PS = Polystyrene
            - Systophome
            - Phenyl groups produce spectral features that distinguish it
            - Hard to recycle and collect
        - PVC = Polyvinyl chloride
            - PVC pipes, flooring, cables, construction materials
            - Contains chlorine atoms
            - Dangerous bc heating releases hydrochen chloride a
            - Needs to be separated
    - Test set with low noise
    - Test set with high noise

What else do I need to learn still?

- Best approaches for hyperspectral ML
- Do a PCA first?
- What channels spike in the different classes?
- Learn about noise in hyperspectral ML
- What kind of ML is best suited

Tasks

- First lets write a python script for viewing the average channel signatures for each category as well as the variance
- Then lets test a number of different ML types and learn what is best for categorically sorting plastic types
    - I can use the lecture slides I wrote for MREN 410 course to choose some ML types
        - Binary thresholds for each channel into weighted voting
            - Use this to identify distinguishing features amongst categories
        - Multifeature regression, linear or logistic - if distinguishing signatures are easy enough to identify
        - High order regressions - if more nuanced method is needed to differentiate
        - Support vector machines or decision trees? Can be used to identify key features like in my bioinformatics assignment
        - Neural network
            - Is it necessary
            - What does it take in terms of compute?
- Does using PCA first help?

Questions to answer in this project

1. Can I identify the category of plastic myself by looking at the NIR spectral patterns?
    a. I.e. Do spikes in certain channels make it easy enough to identify distinguishing characteristics like presence of methyl (CH3) in polypropylene by a human (myself) visually
2. How do different ML models compare in terms of classification accuracy, recall, etc and what does that say about the level of ML model sophistication needed for identifying types of plastics with NIR spectroscopy?

First

- Understand dataset better
    - Does this dataset represent one spectral image?
        - Or are any coordinates repeated?
        - If it is one image, can create a visualization of some sort of shadow of the actual image?
    - Are any channels naturally more prominent (after subtracting dark current signatures), is normalization needed?
- Plot the average spectral patterns with variance for each of the categories
    - Are the categories distinguishable to my eyes at this stage?
    - Can I identify methyl (CH3) for PP, chlorine for PLA, CH2, Ester groups for PET, phenel groups for PS
- Train and test several models:
    - Simple weighted voting
    - Linear regression
    - Logistic regression
    - High order regression
    - Neural network
- Compare and interpret results

This repo contains a machine learning project I conducted individually to further my understanding of near infrared (NIR) spectroscopy and the implementation of machine learning (ML) workflows to analyze NIR hyperspectral imagery. For this project, I used a publically

available labeled NIR hyperspectral dataset to build and compare machine learning frameworks for the task of classifying types of plastic based on NIR signatures.